The Effects of Highly Pathogenic Avian Influenza on Chicken and Turkey Prices

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Econ 740

December 2025

1 Introduction

The 2022 outbreak of Highly Pathogenic Avian Influenza (HPAI) is the largest in American history, resulting in the culling of over 183 million domestic birds since January 2022 (Biondo 2025). U.S. poultry production operates at enormous scale: the egg-laying flock exceeds 378.5 million hens, and processors produced 9.4 billion broiler chickens and 218 million turkeys in 2023 alone (Economic Research Service (ERS) 2025b; Economic Research Service (ERS) 2025a). Egg production took the heaviest blow, as egg-laying hens accounted for 75% of all domestic poultry culled (Biondo 2025). The table-egg-laying hen population fell 5% relative to pre-outbreak years, and the national monthly average egg price peaked at $6.23 per dozen in March 2022 (Biondo 2025).

Avian influenza, commonly called "bird flu", is a contagious viral disease of domestic and wild birds. Few strains can be passed to humans, so its largest effects are economic, falling primarily on the commercial flocks used for egg and poultry production (U.S. Department of Agriculture (USDA) 2025). Poultry meat is also exposed: over 5 million turkeys were culled in the first 7 months of the outbreak, roughly 2.5 percent of all turkey meat production (Economic Research Service (ERS) 2025a). The United States produced over 40 billion pounds of chicken and 30 billion pounds of turkey in 2020, has competitive global advantage in chicken production as a major producer of soybean meal and corn (the primary ingredients of poultry feed), and exports roughly 15-20% of broiler chicken production and 10% of turkey production each year (Economic Research Service (ERS) 2025b). Such a large footprint leaves the industry vulnerable to HPAI, with the potential for inflationary price effects and lost exports. While considerable work documents the effects of HPAI on egg prices, less analysis exists for poultry meat.

Naturally, a sharp drop in food supply puts upward pressure on prices. This paper contributes to two related literatures. The first examines agricultural supply shocks and shows that their effects depend critically on the structure of the agri-food supply chain. Wahdat and Lusk (2022) measure the vulnerability of U.S. food manufacturing industries to upstream industries and labor occupations, showing that shocks to key inputs or labor shortages can propagate strongly through the food system. Using a Cobb–Douglas production framework and an upstream exposure model, they demonstrate that the agri-food sector's position between agricultural production and food retail makes it particularly sensitive to disruptions in intermediate inputs and labor. Complementing this work, Nava, Ridley, and Dall'erba (2023) develop a Ricardian general equilibrium model of U.S. interstate agri-food trade to examine how states absorb adverse production shocks. Their results show that vulnerability largely hinges on a state's ability to replace local production losses through trade with other states, regardless of comparative advantage, and on the state's own productive capacity. Together, these studies show that both vertical integration and spatial trade linkages are central

in determining how agricultural production shocks translate into broader welfare and supply chain impacts.

The second literature examines how infectious disease affects food production and why biosecurity matters. Schulz and Tonsor (2015) study the 2013 outbreak of porcine epidemic diarrhea virus (PEDV) in the United States, which is highly fatal to pigs. Comparing year-over-year swine supply and cost per unit, they descriptively show large revenue losses for pork producers and higher pork prices for consumers. Because the disease spreads from pig to pig (like avian flu), the infected population can grow rapidly, showing how infectious disease raises costs for agri-food producers (Schulz and Tonsor 2015) and underscoring the importance of well-defined biosecurity (Schulz and Tonsor 2015; Pudenz, Schulz, and Tonsor 2019). Similar economic and biosecurity effects appear in American egg production (Mitchell, Thompson, and Malone 2024; Huang, Hagerman, and Bessler 2016).

Mitchell, Thompson, and Malone (2024) study the effect of HPAI on egg and chicken production in 2022-2023, showing that biological lags in infection and stock repopulation lead to misspecification in difference-in-differences models. Under USDA "stamp out" regulations, entire flocks are depopulated if infected or suspected of infection (U.S. Department of Agriculture (USDA) 2025), and egg-laying hens take 4.5-6 months to reach production capacity, creating production lags and cumulative effects for producers. The study also highlights the difficulty of causal analysis of infectious disease: because shocks persist for up to 26 weeks, binary identification underestimates total effects and therefore overestimates consumer surplus. After accounting for biological lags and cumulative producer effects, HPAI infections raise average egg prices by an estimated 5.3% across regions (Mitchell, Thompson, and Malone 2024). Flocks can also be infected by wild fowl and then infect adjacent flocks, again highlighting the importance of biosecurity (Mitchell, Thompson, and Malone 2024), and spread is hard to contain because wild birds pose a constant threat of environmental spillover (Green et al. 2023).

Huang, Hagerman, and Bessler (2016) study the previous HPAI epidemic and estimate that 10% of total egg-laying inventory was lost. Large shocks are generally not localized and tend to affect prices in adjacent regions (Huang, Hagerman, and Bessler 2016). Production shocks in the Midwest, the largest regional chicken and egg producer, affected prices and price discovery elsewhere, while more self-sufficient regions staved off the more harmful effects (Huang, Hagerman, and Bessler 2016). Ajoseh, Thornsbury, and Onel (2025) study HPAI outbreaks from 2012-2024 using Vector Error Correction Models, finding that retail prices account for 39% of the adjustment toward long-term equilibrium, whereas wholesale prices show no evidence of such adjustment. Together, these studies suggest that the price effects of disease shocks depend on both biological production lags and the ability of regional markets to absorb lost supply. Ultimately, viral diseases pose a large and often unpreventable threat to domestic poultry supply.

While most research on HPAI infection research focuses on the egg industry, this study looks at how

HPAI outbreaks affected the prices of chicken and turkey meat. I use USDA-defined regions to account for the structure of regional supply chains. To my knowledge, this is the first study to use biological lags in production among chicken and turkey products as laid out by Mitchell, Thompson, and Malone (2024). Results indicate that poultry prices remained largely resilient to the 2022–2025 HPAI outbreaks, standing in sharp contrast to the dramatic price spikes observed in the egg industry. While the poultry sector incurred large stock losses, the estimated price effects were mostly statistically insignificant and, in several specifications, negative. This unexpected downward pressure on prices suggests that secondary market events, such as international trade restrictions and the subsequent diversion of exports back into the domestic market, substitution effects and inter-regional trade likely offset supply-side shortages. Furthermore, the analysis reveals that frozen chicken products were the primary source of significant inflationary effects, potentially due to their vulnerability to pandemic-related labor shocks and inter-regional trade disruptions. Ultimately, these findings suggest that the shorter biological production cycles for broilers and turkeys, combined with supply chain flexibility, allow the poultry meat market to withstand infectious disease shocks more effectively than the egg sector. The rest of this paper is organized as follows: Section 2 describes the data and primary sources. Section 3 details the methodology, including the construction of the cumulative lag variables (3.1), the inclusion of covariates and controls (3.2), and the empirical strategy (3.3). Section 4 reports the results, followed by a discussion of the potential mechanisms and explanatory pathways in Section 5. Section 6 provides concluding remarks.

2 Data

I construct a region by week panel from February 2022 to November 2025 to study the effects of HPAI on poultry prices. Regions are defined according to USDA Agricultural Marketing Service (AMS) classifications. The panel is unbalanced due to variation in price data from retail advertising activity and weekly COVID-19 pandemic controls, which stop being updated May 2023. Data on HPAI detections are obtained from the USDA Animal and Plant Health Inspection Service (APHIS) Confirmed HPAI Detections dashboard, covering February 2022 through November 2025. Confirmed detections in commercial flocks are aggregated to the region-week level using AMS region definitions. These data provide the primary measure of disease exposure. Retail poultry prices are drawn from the USDA AMS Weekly Retail Feature dataset, which reports advertised retail prices across U.S. grocery outlets. The sample includes all chicken and turkey products with prices reported per pound, classified by species and condition (fresh, frozen, and prepared). For each region and week, average prices per pound are constructed as weighted means for each condition and species, using the number of stores advertising each product as weight, to proxy product prevalence within a region.

Because not all products are advertised every week, price series are incomplete, resulting in an unbalanced panel. Table 1 shows the descriptive statistics of HPAI infections and poultry prices. Tables 1a-6a of the appendix show descriptive statistics for each region.

Table 1: National Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	1,175	3.93	1.22	1.30	14.28
Turkey (USD/lb)	1,101	6.17	1.85	1.17	12.22
Prepared Chicken (USD/lb)	1,159	8.51	1.24	4.23	15.99
Prepared Turkey (USD/lb)	1,095	8.50	1.34	3.68	14.42
Fresh Chicken (USD/lb)	1,152	2.87	0.52	1.30	6.64
Fresh Turkey (USD/lb)	1,079	4.32	0.87	1.40	7.99
Frozen Chicken (USD/lb)	976	2.91	0.96	0.49	19.41
Frozen Turkey (USD/lb)	681	2.24	1.28	0.72	9.99
Affected Chickens (Millions)	1,188	0.12	0.38	0.00	3.90
Affected Turkeys (Millions)	1,188	0.30	0.79	0.00	5.27
Affected Chicken Flocks	1,188	0.79	1.96	0	17
Affected Turkey Flocks	1,188	7.44	18.73	0	118

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding to 10- and 18-week rolling sums for chicken and turkeys, respectively. They do not reflect individual infection events.

Feed price ratio data for chicken and turkey feed are obtained from the USDA Economic Research Service Feed Grains Database (September 2020–July 2025). A feed price-ratio defined as the price per pound of live-weight broilers or turkeys divided by the price per pound of feed, with higher values indicating greater producer profitability. Monthly feed prices are matched to weekly observations by calendar month. Broader farm input costs are captured using the USDA-NASS Prices Paid Index for Commodities and Services Interest, Taxes, and Farm Wage Rates (PPITW). Monthly national trends data for "HPAI" and "bird flu" from Google Trends. These controls are explained further in the next section.

3 Methodology

3.1 HPAI Specification

A key part of this study is accounting for biological lags in repopulating culled flocks of birds. I follow the methodology created in Mitchell, Thompson, and Malone (2024) which outlines how to capture biological lags within the egg-laying industry. Because the egg supply chain does not have excess layer capacity, Mitchell, Thompson, and Malone (2024) first created this specification to account for the time it takes to repopulate a flock of egg-laying hens, which is 26 weeks. Thus, they create 26 week rolling totals for both the number of birds affected and the number of flocks affected, named CBBirds and CBFlocks respectively.

This reflects the cumulative effect of HPAI outbreaks within egg production accounting for biological lags. I utilize a similar process in constructing my HPAI treatment variables. Commercial broiler chicken and turkey production also requires accounting for biological lags, but is shorter than the 26 weeks for egg-laying hens. Commercial poultry production begins from egg, produced by selected breeders, and thus it also does not hold excess capacity. Broiler chickens are generally slaughtered before 2 months, and always under 10 weeks (Economic Research Service (ERS) 2025b). For turkeys it is longer, and are generally slaughtered around 4 months (Economic Research Service (ERS) 2025b). Each poultry type, I construct two variables, CBBirds and CBFlocks, which account for the cumulative rolling total of birds and flocks affected. For 'CBBirds Turkeys' the rolling cumulative total is 18 weeks, and for 'CBBirds Chickens' it is 10 weeks (I refer to these cumulative bioligcal lag variables using 'HPAI' or 'CB' interchangeably). For each confirmed infection unique location, I hold the affected amount constant for the respective amount of weeks and then return to 0, to reflect that the location is 'treated' for the period of time it takes to repopulate that flock. If the same location becomes infected again, the counter is reset and the amount affected reflects the sum of the two amounts. This assumes that all production in one location takes place in the same flock, as I cannot differentiate whether it was the same flock or different flock that was infected, consistent with the methodology in Mitchell, Thompson, and Malone (2024). Breeder birds are excluded from this analysis, as whether the offspring becomes egg-laying hens or broiler is indistinguishable from infection data.

3.2 Covariates and Input Prices

While HPAI outbreaks naturally restrict poultry supply and drive prices upward, the 2022–2023 waves coincided with significant global macroeconomic shocks, namely the COVID-19 pandemic and the Russia–Ukraine conflict. To avoid confounding localized disease disruptions with broad inflationary pressures, I need to control for macroeconomic factors. However, due to the cessation of weekly COVID 19 reporting in 2019, this greatly reduces the sample size. Because of this, I run all model specifications with COVID controls and without, to understand the role COVID macroeconomic shocks played in affecting poultry prices. Furthermore, standard metrics like the CPI are unsuitable here because they are endogenously linked to poultry prices. Similarly, core CPI excludes the very food-price movements essential to this analysis. Following the approach of Mitchell, Thompson, and Malone (2024), I utilize the USDA's Prices Paid Index for Commodities and Services, Interest,Taxes, and Farm Wage Rates (PPITW) as a proxy for broader cost pressures. PPITW is a comprehensive measure of the costs faced by agricultural producers it is composed of five distinct sub-indexes: the Index of Production Items (based on specific price estimates), as well as dedicated indexes for farm interest payments, real estate taxes, hired labor wages, and the costs associated

with family living expenses. This allows the model to isolate non-disease drivers such as labor, feed, and overhead costs from the specific impact of HPAI. I also supplement PPITW with controls for feed price ratios of chicken and turkey. It is possible that HPAI infections reduce demand of chicken and turkey products via consumer fear of contacting HPAI, whether factual or not. To control for this, I include monthly national Google Trends for the terms "HPAI" and "Bird Flu" to proxy consumer awareness. Finally, prices of poultry can also be affected through a substitution effect if the price of substitute goods falls. I include the natural log of weekly pork and beef products, using the same weighting scheme as the outcome variables, to control for such substitution effects. As this is a log-log comparison of prices, the coefficients in these covariates will reflect the cross-price elasticity of pork or beef to chicken or turkey.

3.3 Empirical Strategy

I run a two-way fixed effects model via Ordinary Least Squares for each outcome, restricting the panel by species and COVID-19 controls:

$$Y_{rt} = \beta_0 + \beta_1 \text{HPAI}_{rt} + \delta \mathbf{X}'_{rt} + \alpha_r + \eta_t + \epsilon_{rt} \tag{1}$$

Where Y_{rt} is the natural log of the weighted average price of the species product type in region r at week t, HPAI_{rt} represents either the *CBBirds* or *CBFlocks* count for the corresponding species, β_1 is the coefficient of interest, representing the estimated average effect of a 1 million increase in the rolling cumulative total of birds affected (or number of flocks affected) on the percentage change of regional prices, γ_τ captures the interaction of HPAI with specific weeks to estimate the effect over time, analogous to an event study design, $\mathbf{X}'_{rt}$ is a vector of control variables, α_r and η_t represent region and week fixed effects, respectively. As noted by Cameron, Gelbach, and Miller (2008), a low number of clusters can downward bias standard errors, leading to inaccurate inference. Wild bootstrap standard errors have been shown to be effective when the number of clusters is small (Cameron, Gelbach, and Miller 2008). There are 6 region clusters in my model thus I cluster standard errors by region and apply the wild bootstrap to ensure robust results. This model assumes that the effects of HPAI on prices are linear, and that HPAI shocks are strictly endogenous. It is likely that HPAI effects on rejoice are non-linear, as they may be stronger direct after an outbreak, or increase exponentially due to the cumulative nature of lags. By reporting the average marginal effect over time, it may underestimate large acute shocks and overestimate small ones. Furthermore, HPAI infections may be correlated with wild bird migrational patterns and thus seasons, and this may affect regions differently than others. Region fixed effects will capture time-invariant differences, but seasonality of HPAI infections may affect turkeys (which are sold more in winter) differently than chickens. Week fixed effects will absorb

national seasonal trends affecting turkey prices, however will not capture regional differences in HPAI caused by seasonal migration patterns.

4 Results

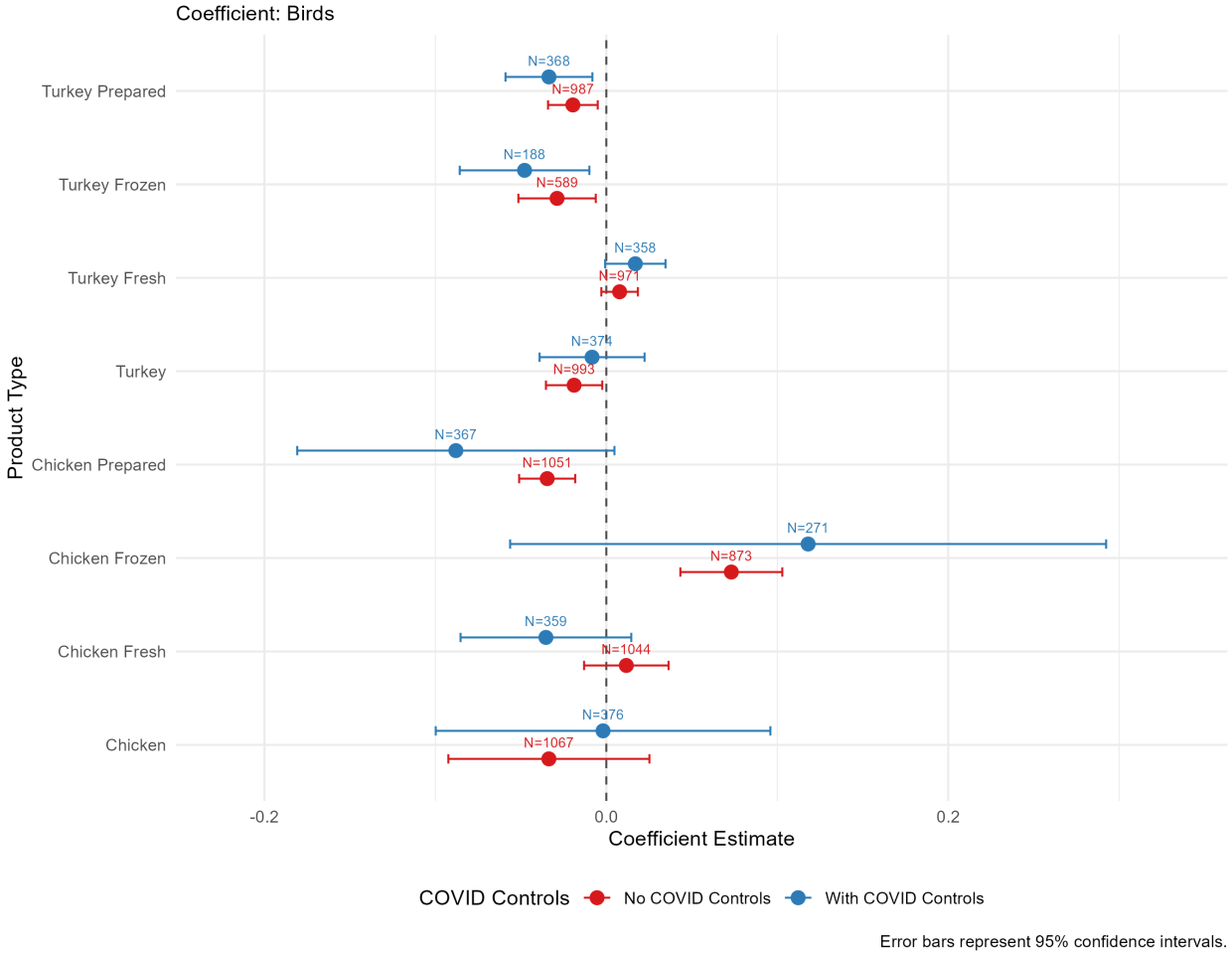
This section reports the results equation 1 analyzing the effect of HPAI outbreaks on poultry prices. Table 2 reports the results of equation 1 on chicken a turkey product price, with and without COVID controls. Each coefficient represents the average effect of HPAI outbreaks in a region on the natural log of the weighted average of all chicken or turkey product prices per lb. Interestingly, average prices for chicken do not seem to be chicken prices seem to be quite resilient to HPAI outbreak shocks. An increase of 1 million chickens affected resulted in an estimated decrease in chicken prices of .2% during COVID years and a decrease in chicken prices of 3.4% overall, without COVID covariates. When measuring HPAI outbreaks by the number of flocks affected, we see an increase in chicken prices of .1% during COVID years and reduction of .6% overall without COVID covariates. They are statistically indistinguishable from 0 and generally relatively small, suggesting that HPAI outbreaks did not significantly economically impact chicken production. The effects on turkey prices are similar, yet more significant when excluding COVID covariates. An increase in 1 million affected turkeys resulted in an average decrease in turkey prices by .8% during COVID years and 1.9% decrease when excluding COVID covariates, significant at a 90% confidence level. When outbreaks are measured in the number of flocks affected, an increase in 1 flock affected resulted in an estimated average decrease in the price of turkey by .03% during COVID years and a .1% decrease when excluding COVID covariates, significant at the 90% confidence level.

Table 2: Effects of HPAI on Chicken and Turkey Products

	Log Price Chicken				Log Price Turkey			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HPAI Birds	-0.002		-0.034					
	p=0.971		p=0.314					
HPAI Flocks		0.001		-0.006				
		p=0.902		p=0.186				
HPAI Birds					-0.008		-0.019*	
					p=0.618		p=0.075	
HPAI Flocks						-0.0003		-0.001*
						p=0.655		p=0.060
COVID Controls	Yes	Yes	No	No	Yes	Yes	No	No
Obs	376	376	1067	1067	374	374	993	993
R2	0.628	0.628	0.369	0.368	0.748	0.748	0.668	0.668

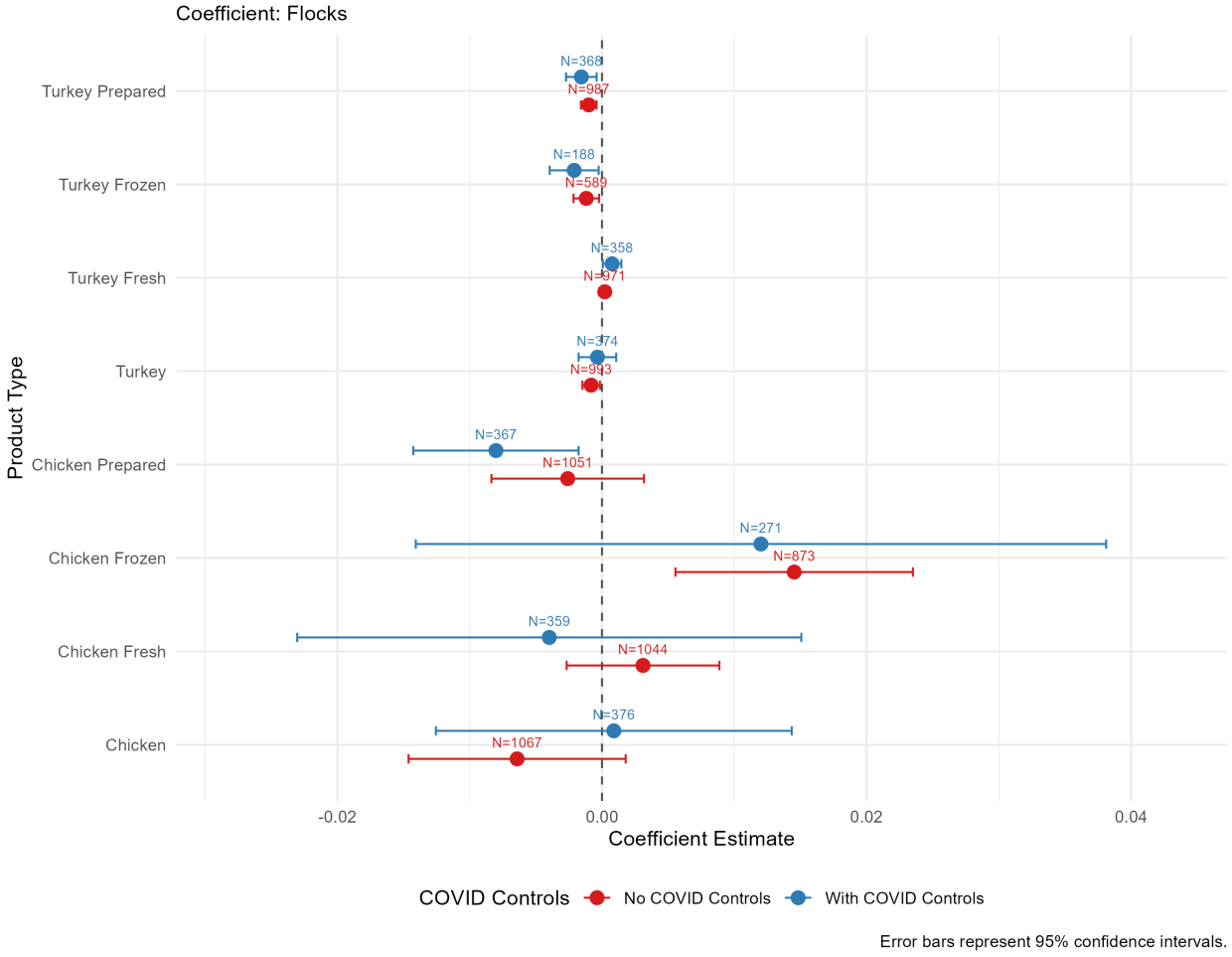
Note: *p<0.1; **p<0.05; ***p<0.01. Birds reported in millions. Standard errors clustered at the region level and wild-bootstrapped. Bootstrapped p-values shown. All specifications include covariates for feed price ratios, PPITW, regional monthly unemployment rates, substitute good prices, and HPAI specific Google Trends terms.

Figure 1: Effects of HPAI Outbreaks on Different Chicken and Turkey Product Types



Figures 1 and 2 report the heterogeneity in average effects on chicken and turkey prices among all product types, for HPAI outbreaks measured in birds and flocks, respectively. When measuring the effects of HPAI outbreaks in birds affected in Figure 1, there are signs of heterogeneous effects. When excluding COVID controls, among chicken products, only prepared chicken products are negative, and are statistically significant at the 95% confidence level. Fresh chicken (never frozen) is positive but indistinguishable from 0, and frozen chicken products saw an increase in price of over 5% which is statistically significant from 0. Among turkey products excluding COVID controls, all are negative and statistically significant ecept for fresh turkey, which is positive and statistically insignificant from 0. When including COVID controls, the sample is massively reduced, by 75% roughly, leading to noisier estimates. However, almost all coefficients are larger in magnitude compared to their non-COVID counter parts suggesting that effects without COVID covariates are masked by confounders in COVID years. The results of Figure 1 illuminate the fact that frozen products were affected most by HPAI, with significantly detectable effects among turkey products.

Figure 2: Effects of HPAI Outbreaks on Different Chicken and Turkey Product Types



When measuring the effects of HPAI outbreaks in number of flocks affected in Figure 2 we see very similar effects to Figure 1, however with larger standard errors for chicken products, especially frozen and fresh products when including COVID controls. This suggests that flocks is a less precise measure of HPAI outbreaks, but only slightly. Trends and magnitudes of coefficients hold across models, suggesting that outbreaks were not concentrated among a few large producers.

5 Discussion

The main specification results on all chicken products are statistically indistinguishable from 0 and generally relatively small, suggesting that HPAI outbreaks generally did not significantly economically impact chicken production. This may be due to broiler chickens having a considerably short biological lag of up to 10 weeks, allowing chicken producers to recover quickly. Most estimates, barring frozen chicken, are negative suggesting that HPAI outbreaks lead to lower prices in chicken products. Similar negative effects are seen

among estimates for turkey, statistically significant at the 95% confidence level for prepared and frozen products. This mechanically could not occur without secondary market events.

There are a few possible mechanisms in which HPAI outbreaks lead to lower prices for turkey and some chicken products. First, it may be the case that foreign importers of US poultry exports reduced their imports due to the prevalence of HPAI in the US. This would force US poultry exporters to sell domestically, resulting in an uncharacteristic increase in the domestic supply and reducing prices.

Second, domestic consumers may have substituted away from poultry toward beef or pork, reducing demand for chicken products. Appendix Table 7a reports cross-price elasticities for chicken and turkey relative to these substitutes, where positive coefficients on beef and pork prices would indicate substitution. For chicken, however, most coefficients are negative and all are statistically indistinguishable from zero. Beef coefficients are positive only when controlling for COVID-19 (0.040 to 0.041) and turn negative without these covariates. The pork elasticity ranges from -0.092 to -0.017 and is not statistically significant, suggesting pork was not used as a substitute. Across all specifications, p-values exceed 0.50, indicating that beef and pork price changes had no measurable impact on poultry prices during the sample period. Although beef price changes were larger in magnitude than chicken price reductions, suggesting potential substitution from February 2022 to May 2023, the lack of statistical significance suggests these cross-price effects were likely not a primary driver of poultry price movements.Substitution appears stronger for turkey, though it remains statistically insignificant. Unlike the chicken specifications, all turkey cross-price elasticities are positive, suggesting some substitution: the beef elasticity rises from 0.033 in COVID-controlled models to 0.141 without COVID covariates, while the pork elasticity stays between 0.086 and 0.105. These magnitudes exceed those for chicken but are statistically indistinguishable from zero, so the substitution effects appear weak. Turkey prices therefore move more closely with beef and pork prices than chicken prices do, but substitute prices were not a primary driver of the poultry price reductions observed during the HPAI outbreak. This is consistent with the longer biological lag needed to restore depopulated turkey flocks, which produces larger cumulative outbreak effects. Turkey, along with egg-laying hens, were also the most affected by HPAI outbreaks (Biondo 2025).

Third, Nava, Ridley, and Dall'erba (2023) show that a regions vulnerability to supply shocks is largely explained by their inter-regional trade abilities. Regions that were best able to weather shocks were those who traded with other regions to supplement their supply. Among poultry products, frozen items have the longest shelf life, and would likely be most viable to transport to other regions. While supplementing one's own regional supply with imported frozen poultry would likely not reduce prices below the market clearing rate, it could amplify marginal substitution effects, exerting downward pressure on domestic prices. It is likely that price-reducing effect of HPAI is explained in some part by all three of these potential mechanisms.

This paper does not analyze international or domestic inter-regional trade patterns, and further analysis of trade behaviors during this period may illuminate as to why HPAI outbreaks largely did not affect poultry prices.

There is also evidence that COVID-19 had effects on prices. Among chicken prices, including COVID controls reduced the negative price effects, suggesting it had inflationary pressure on prices. Similar effects are seen among turkey products. Beyond the supply disruptions caused by HPAI, it is possible that the 2022–2023 poultry market was shaped by significant labor and demand shocks stemming from COVID-19. The poultry industry may have faced labor shortages, especially in highly processed frozen products made in factory-like facilities. These dense processing locations were likely more vulnerable to COVID-19 spread and acute labor shortages, which could have raised marginal processing costs and limited the throughput of retail-ready frozen products. Simultaneously, pandemic-induced shifts in consumer behavior may have created a sustained demand shock for frozen chicken; it is possible that consumers prioritized these items for their durability and "stockpiling" potential during market uncertainty. Furthermore, decreased household incomes during COVID-19 may have driven a substitution away from food-away-from-home to lower-cost frozen chicken.

These results offer a contrasting story to the significant impacts that HPAI had on egg prices, as Mitchell, Thompson, and Malone (2024) showed significant positive price effects. With largely negative and statistically insignificant effects, this study shows that poultry products were far less impacted by HPAI outbreaks, despite considerable losses to turkey and chicken flocks, and that market dynamics related to COVID-19 had significant inflationary impacts, that in some cases completely negated negative impacts shown in non-COVID models. It is plausible that shorter biological lags for broiler chickens and turkey products compared to egg-laying hens allowed for the effects of HPAI on prices to be largely insignificant.

6 Conclusion

This study analyzes the 2022–2025 Highly Pathogenic Avian Influenza (HPAI) outbreak's impact on U.S. chicken and turkey markets. By accounting for biological re-population lags and COVID-19 macroeconomic volatility, the results contrast with existing literature on egg prices. While HPAI caused historic losses, price effects on poultry meat were largely insignificant or negative, with significant negative effects appearing specifically in turkey products. This downward pressure suggests secondary market mechanisms, such as export-diversion effects where international trade bans forced surplus poultry into the domestic market. Frozen chicken was the sole source of significant inflationary effects, likely due to its susceptibility to inter-regional trade disruptions and pandemic-related labor shocks. Overall, the poultry meat market ap-

pears more resilient to infectious disease than the egg industry, likely due to shorter biological production cycles and responsive supply-side adjustments. The inclusion of COVID-19 controls further reveals that pandemic-era labor and demand shocks exerted inflationary pressures that masked the isolated impact of HPAI.Furthermore, the lack of significant cross-price elasticity suggests that poultry prices remained decoupled from red meat substitutes during this period. These findings underscore that the economic burden of animal disease is non-uniform across agricultural sectors; supply chain flexibility and shorter growth cycles can mitigate consumer-level impacts. Future research should integrate international trade and inter-regional frozen inventory flows to precisely identify the mechanisms that stabilized the market despite unprecedented biological losses.

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7 Appendix

Table 1a: Regional Summary Statistics: Midwest

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	196	4.28	0.93	2.77	9.97
Turkey (USD/lb)	184	6.36	1.58	1.48	8.87
Prepared Chicken (USD/lb)	196	8.46	0.70	6.92	9.97
Prepared Turkey (USD/lb)	184	8.74	0.79	7.05	10.63
Fresh Chicken (USD/lb)	192	3.00	0.37	2.21	4.01
Fresh Turkey (USD/lb)	184	4.29	0.72	1.91	5.61
Frozen Chicken (USD/lb)	183	3.10	0.71	1.47	6.88
Frozen Turkey (USD/lb)	173	2.27	1.04	0.96	5.70
Affected Chickens (Millions)	198	0.08	0.25	0.00	1.26
Affected Turkeys (Millions)	198	1.47	1.40	0.00	5.27
Affected Chicken Flocks	198	0.30	0.77	0	4
Affected Turkey Flocks	198	34.91	33.25	0	118

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding 10 and 18 weekly rolling sums for chicken and turkeys, respectively. They do no reflect individual infection events.

Table 2a: Regional Summary Statistics: Northeast

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	196	4.08	0.93	2.73	9.91
Turkey (USD/lb)	184	6.43	1.34	2.27	8.45
Prepared Chicken (USD/lb)	196	8.97	0.69	6.55	11.07
Prepared Turkey (USD/lb)	184	8.90	0.61	7.31	10.51
Fresh Chicken (USD/lb)	192	2.92	0.32	2.16	3.80
Fresh Turkey (USD/lb)	184	4.74	0.65	2.67	6.56
Frozen Chicken (USD/lb)	191	2.68	0.46	1.89	4.71
Frozen Turkey (USD/lb)	136	2.20	1.01	0.79	5.55
Affected Chickens (Millions)	198	0.10	0.28	0.00	1.42
Affected Turkeys (Millions)	198	0.02	0.04	0.00	0.13
Affected Chicken Flocks	198	0.86	2.19	0	11
Affected Turkey Flocks	198	1.09	2.26	0	8

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding 10 and 18 weekly rolling sums for chicken and turkeys, respectively. They do no reflect individual infection events.

Table 3a: Regional Summary Statistics: Northwest

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	195	3.85	1.46	1.30	11.99
Turkey (USD/lb)	181	6.93	2.36	1.98	12.22
Prepared Chicken (USD/lb)	179	8.32	1.41	5.32	12.99
Prepared Turkey (USD/lb)	175	9.30	1.48	5.76	14.42
Fresh Chicken (USD/lb)	192	2.85	0.71	1.30	6.64
Fresh Turkey (USD/lb)	162	4.29	1.06	1.40	7.80
Frozen Chicken (USD/lb)	122	3.17	1.24	0.49	10.99
Frozen Turkey (USD/lb)	69	2.26	1.56	0.96	9.99
Affected Chickens (Millions)	198	0.04	0.17	0.00	0.75
Affected Turkeys (Millions)	198	0.00	0.00	0	0
Affected Chicken Flocks	198	0.10	0.44	0	2
Affected Turkey Flocks	198	0.00	0.00	0	0

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding 10 and 18 weekly rolling sums for chicken and turkeys, respectively. They do no reflect individual infection events.

Table 4a: Regional Summary Statistics: Southcentral

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	196	3.71	1.07	2.06	10.74
Turkey (USD/lb)	184	6.14	1.77	1.57	11.01
Prepared Chicken (USD/lb)	196	8.03	1.01	5.64	11.49
Prepared Turkey (USD/lb)	184	7.80	1.41	4.17	11.64
Fresh Chicken (USD/lb)	192	2.66	0.39	1.85	4.19
Fresh Turkey (USD/lb)	184	4.37	0.79	2.04	6.38
Frozen Chicken (USD/lb)	180	2.71	0.76	0.98	7.69
Frozen Turkey (USD/lb)	100	2.54	1.58	0.99	8.98
Affected Chickens (Millions)	198	0.07	0.12	0.00	0.47
Affected Turkeys (Millions)	198	0.07	0.13	0.00	0.49
Affected Chicken Flocks	198	0.96	1.46	0	6
Affected Turkey Flocks	198	2.82	4.70	0	18

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding 10 and 18 weekly rolling sums for chicken and turkeys, respectively. They do no reflect individual infection events.

Table 5a: Regional Summary Statistics: Southeast

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	196	3.95	1.30	2.01	12.53
Turkey (USD/lb)	184	5.46	1.83	1.17	11.08
Prepared Chicken (USD/lb)	196	8.59	1.38	5.75	12.53
Prepared Turkey (USD/lb)	184	8.56	1.20	4.68	12.77
Fresh Chicken (USD/lb)	192	2.97	0.57	1.69	4.24
Fresh Turkey (USD/lb)	184	4.03	0.77	1.65	6.20
Frozen Chicken (USD/lb)	180	2.88	0.71	0.99	5.04
Frozen Turkey (USD/lb)	131	2.10	1.22	0.79	8.51
Affected Chickens (Millions)	198	0.10	0.18	0.00	0.80
Affected Turkeys (Millions)	198	0.02	0.03	0.00	0.11
Affected Chicken Flocks	198	0.93	1.40	0	6
Affected Turkey Flocks	198	1.10	1.75	0	6

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding 10 and 18 weekly rolling sums for chicken and turkeys, respectively. They do no reflect individual infection events.

Table 6a: Regional Summary Statistics: Southwest

Statistic	N	Mean	St. Dev.	Min	Max
Chicken (USD/lb)	196	3.73	1.41	1.98	14.28
Turkey (USD/lb)	184	5.74	1.75	1.39	9.78
Prepared Chicken (USD/lb)	196	8.67	1.71	4.23	15.99
Prepared Turkey (USD/lb)	184	7.74	1.51	3.68	11.18
Fresh Chicken (USD/lb)	192	2.84	0.56	1.71	4.37
Fresh Turkey (USD/lb)	181	4.18	1.03	1.69	7.99
Frozen Chicken (USD/lb)	120	3.09	1.74	1.43	19.41
Frozen Turkey (USD/lb)	72	2.03	1.54	0.72	9.59
Affected Chickens (Millions)	198	0.32	0.78	0.00	3.90
Affected Turkeys (Millions)	198	0.21	0.33	0.00	1.20
Affected Chicken Flocks	198	1.57	3.48	0	17
Affected Turkey Flocks	198	4.70	7.36	0	25

Notes: Prices reflect weighted average price per lb. Affected variables reflect constructed CBBirds and CBFlocks variables, corresponding 10 and 18 weekly rolling sums for chicken and turkeys, respectively. They do no reflect individual infection events.

Table 7a: Effects of HPAI on Chicken and Turkey Products

	Log Price Chicken				Log Price Turkey			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HPAI Birds	−0.002		−0.034					
	p = 0.971		p = 0.314					
HPAI Flocks		0.001		−0.006				
		p = 0.902		p = 0.186				
HPAI Birds					−0.008		−0.019*	
					p = 0.618		p = 0.075*	
HPAI Flocks						−0.0003		−0.001*
						p = 0.655		p = 0.060*
Log Price Beef	0.041	0.040	−0.026	−0.030	0.033	0.033	0.141	0.141
	p = 0.621	p = 0.626	p = 0.826	p = 0.794	p = 0.907	p = 0.908	p = 0.155	p = 0.148
Log Price Pork	−0.092	−0.092	−0.017	−0.021	0.105	0.105	0.086	0.087
	p = 0.521	p = 0.520	p = 0.832	p = 0.796	p = 0.176	p = 0.179	p = 0.283	p = 0.269
COVID Controls	Yes	Yes	No	No	Yes	Yes	No	No
Observations	376	376	1,067	1,067	374	374	993	993
R^2	0.628	0.628	0.369	0.368	0.748	0.748	0.668	0.668

Note: *p<0.1; **p<0.05; ***p<0.01. Birds reported in millions. Standard errors clustered at the region level and wild-bootstrapped. Bootstrapped p-values shown. COVID covariates include weekly new cases and new deaths. All specifications include covariates for feed price ratios, PPITW, regional monthly unemployment rates, substitute good prices, and HPAI specific Google Trends terms.